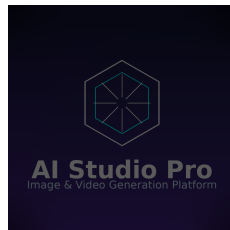
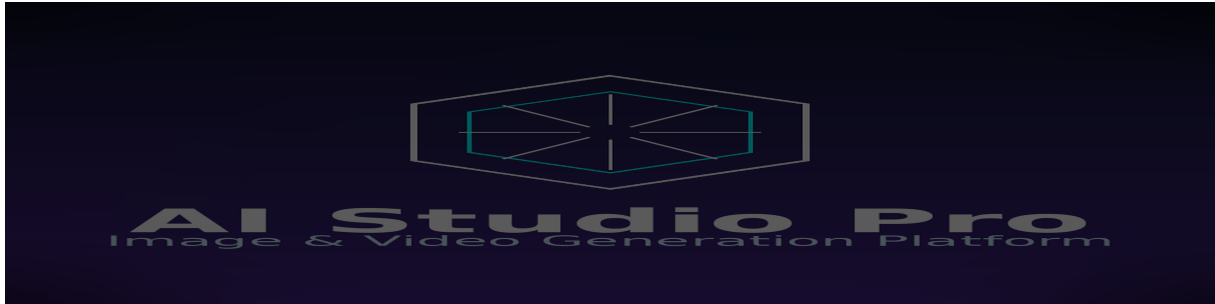


AI Studio Pro — Technical Documentation (Production-Ready)



This document summarizes the architecture, security posture, database design, API surface, and testing strategy for the AI Studio Pro platform (PFE edition).

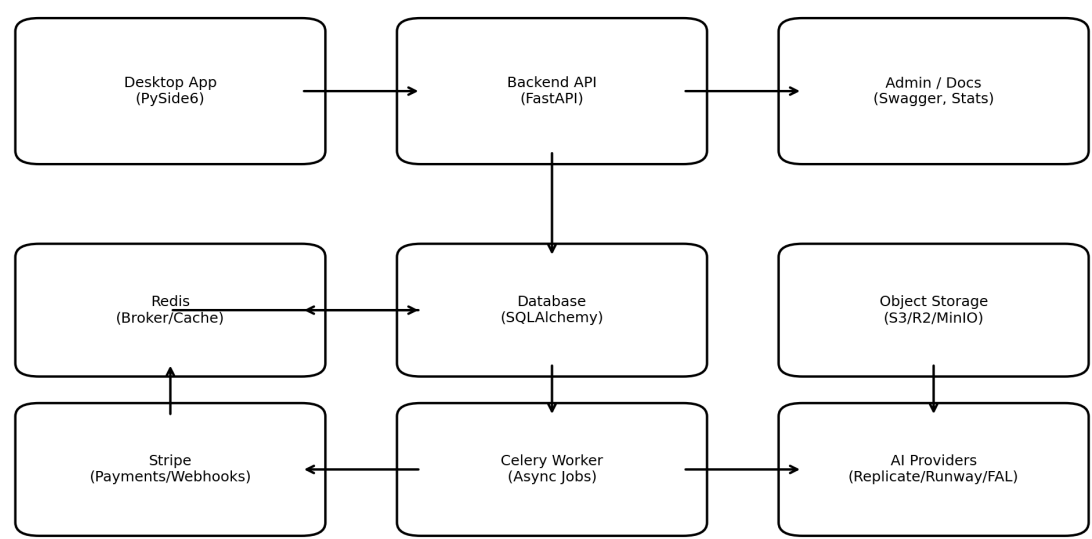
1. Product Overview

AI Studio Pro is a professional AI image/video generation platform with authentication, credits, subscriptions, Stripe payments, background workers (Celery), and multi-client support (Web + Desktop).

Target: a production-ready MVP that can be deployed behind HTTPS with proper secrets management and safe webhook processing.

2. High-Level Architecture

AI Studio Pro — High-Level Architecture

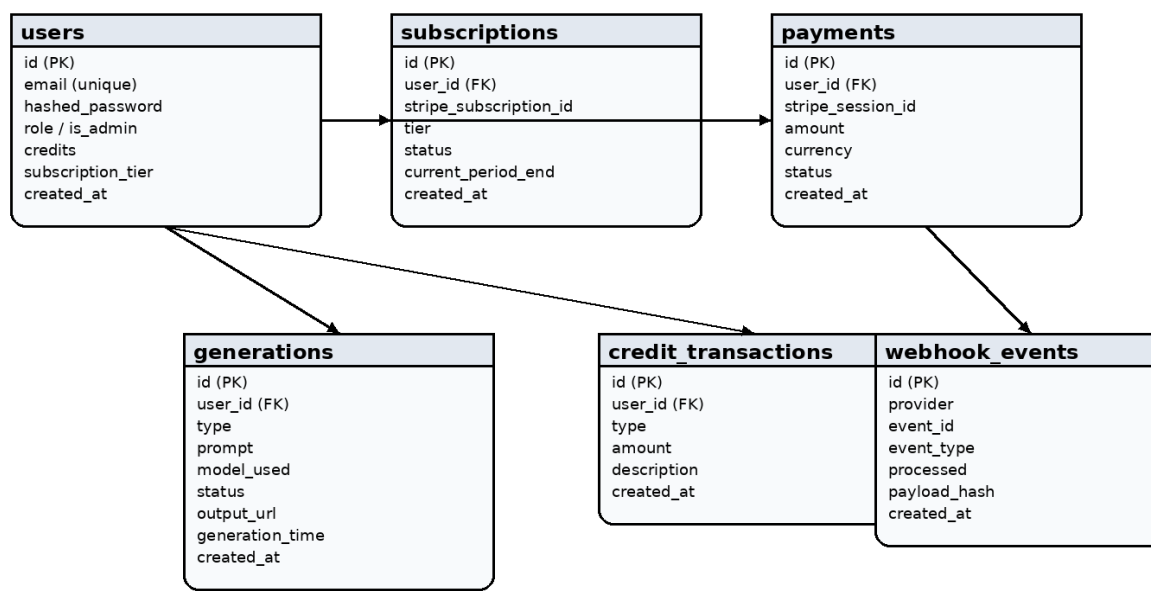


Core components: FastAPI backend, PostgreSQL/SQLite DB (async SQLAlchemy), Redis for caching/rate-limiting support, Celery workers for long jobs, Stripe for billing, Next.js webapp, and a desktop client.

3. Database Design

Main entities: users, subscriptions, payments, generations, credit_transactions, webhook_events.

AI Studio Pro - Database ER Diagram



4. Security & Hardening

- Secrets are loaded from environment variables only (no hardcoded keys).
- Stripe webhooks verify signatures using `STRIPE_WEBHOOK_SECRET` and track processed event IDs.
- JWT access tokens include expiry; refresh tokens are supported; token revocation is available (JTI).
- Security headers middleware enables X-Frame-Options, X-Content-Type-Options, Referrer-Policy, and optional HSTS in production.
- CORS origins are configured via settings; avoid wildcard '*' in production.
- Background jobs run in Celery to avoid request timeouts; DB sessions are handled safely for workers.

5. API Surface (Summary)

Area	Endpoints (base /api/v1)
Auth	POST /auth/register, POST /auth/login, POST /auth/refresh, POST /auth/logout
Users	GET /users/me, PATCH /users/me
Credits	GET /credits/balance, GET /credits/packages, POST /credits/purchase, GET /credits/...
Subscriptions	GET /subscriptions/current, POST /subscriptions/checkout, POST /subscriptions/po...
Generations	POST /generations, GET /generations, GET /generations/{id}
Admin	GET /admin/overview, /admin/generations_per_day, /admin/revenue, /admin/top_...
Webhooks	POST /webhooks/stripe

6. Testing

Backend includes a pytest suite with async TestClient coverage for: health check, auth register/login, and credits endpoints.

Run:

```
cd backend && pip install -r requirements.txt && pytest -q
```

7. Deployment Notes

Recommended: Docker Compose in production with PostgreSQL + Redis, and a reverse proxy (Nginx/Caddy) providing HTTPS.

See docs/PROD_CHECKLIST.md for the full checklist.